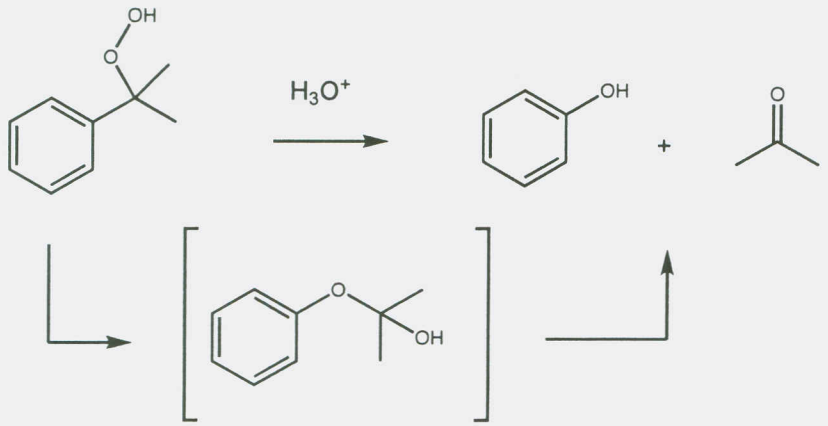
Organic Cume April 2, 2009

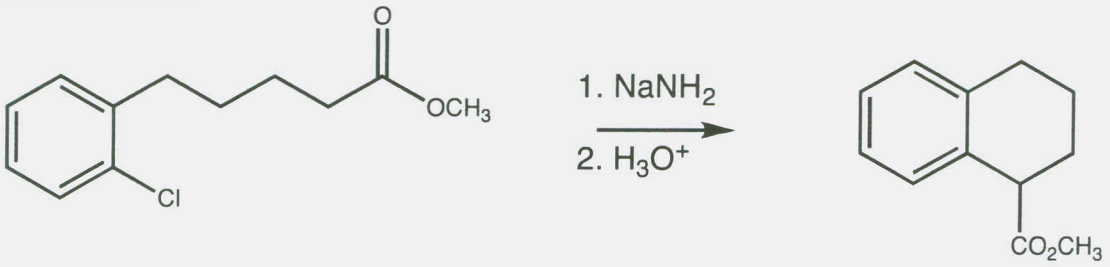
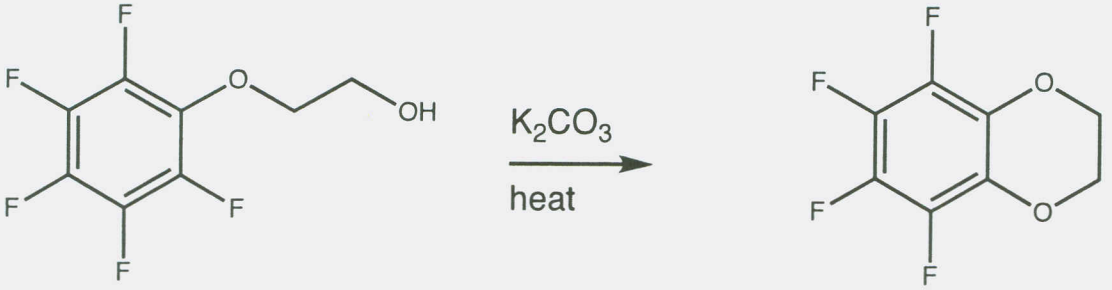
The following problems are all taken from Carey's Organic Chemistry, 7th Edition.

1. Problem 24.29 The industrial preparation of phenol proceeds via acid catalyzed decomposition of cumene hydroperoxide. Propose a step-by-step reaction mechanism for the conversion of cumene hydroperoxide to the hemi-acetal shown. You don't need to show the mechanism for the conversion of the hemi-acetal to phenol and acetone.



2. Draw the step-by-step reaction mechanism for the following reactions.

Problem 23.28b



3. Problem 23.28d